Lecture Notes: Tensor Calculus

Based on Youtube series by eigenchris

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1 Multi-variable Calculus Review

For this lecture notes on Tensor Calculus, there are some concepts and notes that we need to quickly review on multi-variable calculus, which need to be clear and understood before proceeding.
Here we'll give a brief review of the following topics:

- Single-variable Derivatives and chain rule;
- Single-variable Integrals and differentials (change of variable);
- Multi-variable Partial derivatives and chain rule;
- Gradient of a function;
- Directional Derivatives
- Multi-variable Integrals and differentials;
- Arch length, velocity vector tangent to a curve.

1.1 Single-Variable Derivatives and Chain Rule

The derivative of a function $f(x)$ with respect to x measures the instantaneous rate of change:

$$\frac{\mathrm{d}f}{\mathrm{d}x} = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$$

Basic derivative rules:

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}x}(x^n) &= nx^{n-1} \\
\frac{\mathrm{d}}{\mathrm{d}x}(e^x) &= e^x \\
\frac{\mathrm{d}}{\mathrm{d}x}(\sin x) &= \cos x \\
\frac{\mathrm{d}}{\mathrm{d}x}(\cos x) &= -\sin x \\
\frac{\mathrm{d}}{\mathrm{d}x}(\ln x) &= \frac{1}{x}
\end{aligned}$$

Sum and product rules:

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}x}(f+g) &= \frac{\mathrm{d}f}{\mathrm{d}x} + \frac{\mathrm{d}g}{\mathrm{d}x} \\
\frac{\mathrm{d}}{\mathrm{d}x}(fg) &= f\frac{\mathrm{d}g}{\mathrm{d}x} + g\frac{\mathrm{d}f}{\mathrm{d}x}
\end{aligned}$$

Chain rule: For a composition of functions $f(g(x))$:

$$\frac{\mathrm{d}}{\mathrm{d}x} f(g(x)) = \frac{\mathrm{d}f}{\mathrm{d}g} \cdot \frac{\mathrm{d}g}{\mathrm{d}x}$$

More explicitly, if $y = f(u)$ and $u = g(x)$, then:

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}y}{\mathrm{d}u} \cdot \frac{\mathrm{d}u}{\mathrm{d}x}$$

Example: Let $y = \sin(x^2)$. Setting $u = x^2$, we have $y = \sin(u)$:

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \cos(u) \cdot 2x = 2x\cos\left(x^2\right)$$

1.2 Single-Variable Integrals and Change of Variable

The definite integral of $f(x)$ from a to b represents the signed area under the curve:

$$\int_a^b f(x)\,dx$$

Fundamental Theorem of Calculus: If $F(x)$ is an antiderivative of $f(x)$ (i.e., $\frac{\mathrm{d}F}{\mathrm{d}x} = f$), then:

$$\int_a^b f(x)\,dx = F(b) - F(a)$$

Basic integrals:

$$\int x^n\,dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$$

$$\int \frac{1}{x}\,dx = \ln|x| + C$$

$$\int e^x\,dx = e^x + C$$

$$\int \sin x\,dx = -\cos x + C$$

$$\int \cos x\,dx = \sin x + C$$

Change of Variable (Substitution): To integrate $f(g(x))g'(x)$, substitute $u = g(x)$, so that $du = g'(x)\,dx$:

$$\int f(g(x))g'(x)\,dx = \int f(u)\,du$$

For definite integrals, the limits change accordingly: if $u = g(x)$, then $x = a \to u = g(a)$ and $x = b \to u = g(b)$.

Example: Evaluate $\int x\cos(x^2)\,dx$. Let $u = x^2$, then $du = 2x\,dx$, so $x\,dx = \frac{1}{2}du$:

$$\int x\cos\left(x^2\right)dx = \int \cos(u) \cdot \frac{1}{2}\,du = \frac{1}{2}\sin(u) + C = \frac{1}{2}\sin\left(x^2\right) + C$$

1.3 Multi-Variable Partial Derivatives and Chain Rule

For a function $f(x, y)$ of multiple variables, the **partial derivative** with respect to one variable treats all other variables as constants.

Notation:

$$\frac{\partial f}{\partial x}, \quad \frac{\partial f}{\partial y}, \quad f_x, \quad f_y$$

Definition:

$$\frac{\partial f}{\partial x} = \lim_{h \to 0} \frac{f(x+h, y) - f(x, y)}{h}$$

Example: For $f(x, y) = x^2 y + \sin(xy)$:

$$\frac{\partial f}{\partial x} = 2xy + y\cos(xy)$$
$$\frac{\partial f}{\partial y} = x^2 + x\cos(xy)$$

Multivariable Chain Rule: Consider $z = f(x, y)$ where $x = x(t)$ and $y = y(t)$ are functions of t. Then:

$$\frac{\mathrm{d}z}{\mathrm{d}t} = \frac{\partial f}{\partial x}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{\partial f}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}t}$$

More generally, if $z = f(x, y)$ and $x = x(u, v)$, $y = y(u, v)$, then:

$$\frac{\partial z}{\partial u} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial u}$$

Example: Let $z = x^2 + y^2$ with $x = r\cos\theta$ and $y = r\sin\theta$. Then:

$$\begin{aligned}\frac{\partial z}{\partial r} &= \frac{\partial z}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial r} \\ &= (2x)(\cos\theta) + (2y)(\sin\theta) \\ &= 2r\cos^2\theta + 2r\sin^2\theta = 2r\end{aligned}$$

Einstein Summation Convention: Using indexed notation where x^i represents the i-th coordinate, the chain rule can be written compactly as:

$$\frac{\partial f}{\partial u^j} = \frac{\partial f}{\partial x^i}\frac{\partial x^i}{\partial u^j}$$

where summation over the repeated index i is implied (no explicit $\sum$ symbol needed). This convention will be used extensively in tensor calculus.

1.4 Gradient of a Function

The **gradient** of a scalar function $f(x, y, z)$ is a vector field that points in the direction of greatest rate of increase of the function.

Definition: For a function $f : \mathbb{R}^n \to \mathbb{R}$, the gradient is:

$$\boldsymbol{\nabla} f = \nabla f = \begin{pmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{pmatrix}$$

In three dimensions:

$$\boldsymbol{\nabla} f = \frac{\partial f}{\partial x}\hat{\imath} + \frac{\partial f}{\partial y}\hat{\jmath} + \frac{\partial f}{\partial z}\hat{k}$$

Geometric interpretation:

- $\boldsymbol{\nabla} f$ points in the direction of steepest ascent of f
- $|\boldsymbol{\nabla} f|$ gives the rate of change in that direction
- $\boldsymbol{\nabla} f$ is perpendicular to level curves (2D) or level surfaces (3D) of f

Example: For $f(x, y) = x^2 + y^2$:

$$\boldsymbol{\nabla} f = 2x\hat{\imath} + 2y\hat{\jmath} = 2(x\hat{\imath} + y\hat{\jmath})$$

At point $(1, 1)$: $\boldsymbol{\nabla} f = 2\hat{\imath} + 2\hat{\jmath}$, pointing radially outward.

Applications:

- *Optimization*: Finding maxima/minima (critical points where $\boldsymbol{\nabla} f = 0$)
- *Physics*: Conservative forces $\vec{F} = -\boldsymbol{\nabla} U$ (gravitational, electric fields)
- *Machine learning*: Gradient descent for minimizing loss functions
- *Fluid dynamics*: Pressure gradients drive fluid flow

1.5 Directional Derivatives

While partial derivatives measure the rate of change along coordinate axes, the **directional derivative** measures how fast a function changes in an arbitrary direction.

Definition: The directional derivative of f at point $\vec{a}$ in the direction of unit vector $\vec{u}$ is:

$$D_{\vec{u}} f(\vec{a}) = \lim_{h \to 0} \frac{f(\vec{a} + h\vec{u}) - f(\vec{a})}{h}$$

This gives the instantaneous rate of change of f as we move from $\vec{a}$ in direction $\vec{u}$.

Key requirement: $\vec{u}$ must be a *unit vector* (i.e., $|\vec{u}| = 1$) for the directional derivative to represent the true rate of change per unit distance.

Formula using gradient: The directional derivative can be computed using the dot product:

$$D_{\vec{u}} f = \boldsymbol{\nabla} f \cdot \vec{u} = |\boldsymbol{\nabla} f||\vec{u}| \cos\theta = |\boldsymbol{\nabla} f| \cos\theta$$

where θ is the angle between $\boldsymbol{\nabla} f$ and $\vec{u}$.

Important properties:

- **Maximum rate:** $D_{\vec{u}}f$ is maximized when $\vec{u}$ points in the direction of $\boldsymbol{\nabla} f$ (i.e., $\theta = 0$):

$$\max_{\vec{u}} D_{\vec{u}}f = |\boldsymbol{\nabla} f|$$

- **Minimum rate:** $D_{\vec{u}}f$ is minimized when $\vec{u}$ points opposite to $\boldsymbol{\nabla} f$ (i.e., $\theta = \pi$):

$$\min_{\vec{u}} D_{\vec{u}}f = -|\boldsymbol{\nabla} f|$$

- **Zero rate:** $D_{\vec{u}}f = 0$ when $\vec{u}$ is perpendicular to $\boldsymbol{\nabla} f$ (i.e., $\theta = \pi/2$), meaning along level curves/surfaces

Example 1: Find the directional derivative of $f(x, y) = x^2 + xy$ at $(1, 2)$ in the direction toward $(4, 6)$.

First, compute the gradient:

$$\boldsymbol{\nabla} f = (2x + y)\hat{\imath} + x\hat{\jmath} \quad \Rightarrow \quad \boldsymbol{\nabla} f(1, 2) = 4\hat{\imath} + \hat{\jmath}$$

The direction vector from $(1, 2)$ to $(4, 6)$ is $\vec{v} = 3\hat{\imath} + 4\hat{\jmath}$. Normalize it:

$$\vec{u} = \frac{\vec{v}}{|\vec{v}|} = \frac{3\hat{\imath} + 4\hat{\jmath}}{\sqrt{9 + 16}} = \frac{3}{5}\hat{\imath} + \frac{4}{5}\hat{\jmath}$$

The directional derivative is:

$$D_{\vec{u}}f = \boldsymbol{\nabla} f \cdot \vec{u} = (4\hat{\imath} + \hat{\jmath}) \cdot \left(\frac{3}{5}\hat{\imath} + \frac{4}{5}\hat{\jmath}\right) = \frac{12}{5} + \frac{4}{5} = \frac{16}{5}$$

Example 2: For $f(x, y) = x^2 + y^2$ at $(3, 4)$, find the direction of maximum increase and the maximum rate.

$$\boldsymbol{\nabla} f = 2x\hat{\imath} + 2y\hat{\jmath} \quad \Rightarrow \quad \boldsymbol{\nabla} f(3, 4) = 6\hat{\imath} + 8\hat{\jmath}$$

The direction of maximum increase is along $\boldsymbol{\nabla} f$. The unit vector is:

$$\vec{u}_{\max} = \frac{6\hat{\imath} + 8\hat{\jmath}}{\sqrt{36 + 64}} = \frac{6\hat{\imath} + 8\hat{\jmath}}{10} = \frac{3}{5}\hat{\imath} + \frac{4}{5}\hat{\jmath}$$

The maximum rate of increase is:

$$|\boldsymbol{\nabla} f(3, 4)| = \sqrt{36 + 64} = 10$$

1.6 Multi-Variable Integrals and Differentials

Double and triple integrals extend the concept of integration to functions of multiple variables, computing volumes under surfaces or higher-dimensional analogues.

Double integral: For a function $f(x, y)$ over region R:

$$\iint_R f(x, y)\, dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y)\, dy\, dx$$

where $dA = dx\,dy$ is the area element.

Triple integral: For a function $f(x, y, z)$ over region V:

$$\iiint_V f(x, y, z)\,dV$$

where $dV = dx\,dy\,dz$ is the volume element.

Change of Variables (Jacobian): When transforming coordinates $(x, y) \to (u, v)$, the area element transforms according to:

$$dx\,dy = |J|\,du\,dv$$

where J is the **Jacobian determinant**:

$$J = \frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \frac{\partial x}{\partial u}\frac{\partial y}{\partial v} - \frac{\partial x}{\partial v}\frac{\partial y}{\partial u}$$

For three dimensions $(x, y, z) \to (u, v, w)$:

$$J = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{vmatrix}$$

Example - Polar coordinates: For $x = r\cos\theta$, $y = r\sin\theta$:

$$J = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} = r\cos^2\theta + r\sin^2\theta = r$$

Therefore: $dx\,dy = r\,dr\,d\theta$

This explains why integrals in polar coordinates have the extra factor of r:

$$\iint_R f(x, y)\,dx\,dy = \iint_{R'} f(r\cos\theta, r\sin\theta)\,r\,dr\,d\theta$$

Differential forms: A differential df of a function $f(x, y)$ is:

$$df = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy$$

This represents the infinitesimal change in f due to infinitesimal changes dx and dy in the coordinates. More generally:

$$df = \boldsymbol{\nabla} f \cdot d\vec{r}$$

where $d\vec{r} = dx\hat{i} + dy\hat{j} + dz\hat{k}$ is the displacement vector.

Einstein notation: Using indexed coordinates x^i, the differential can be written compactly as:

$$df = \frac{\partial f}{\partial x^i}dx^i$$

where summation over repeated index i is implied. This notation emphasizes that the differential is a linear combination of coordinate differentials weighted by partial derivatives.

1.7 Arc Length and Velocity Vector Tangent to a Curve

A **parametric curve** in space is described by a position vector $\vec{r}(t)$ that depends on parameter t:

$$\vec{r}(t) = x(t)\hat{\imath} + y(t)\hat{\jmath} + z(t)\hat{k}$$

Velocity vector: The derivative of position with respect to the parameter gives the velocity:

$$\vec{v}(t) = \frac{\mathrm{d}\vec{r}}{\mathrm{d}t} = \frac{\mathrm{d}x}{\mathrm{d}t}\hat{\imath} + \frac{\mathrm{d}y}{\mathrm{d}t}\hat{\jmath} + \frac{\mathrm{d}z}{\mathrm{d}t}\hat{k} = \dot{x}\hat{\imath} + \dot{y}\hat{\jmath} + \dot{z}\hat{k}$$

The velocity vector $\vec{v}(t)$ is **tangent to the curve** at each point and points in the direction of motion.

Speed: The magnitude of the velocity vector gives the speed:

$$|\vec{v}(t)| = \sqrt{\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)^2 + \left(\frac{\mathrm{d}y}{\mathrm{d}t}\right)^2 + \left(\frac{\mathrm{d}z}{\mathrm{d}t}\right)^2}$$

Unit tangent vector: Normalizing the velocity gives a unit vector tangent to the curve:

$$\vec{T}(t) = \frac{\vec{v}(t)}{|\vec{v}(t)|}$$

Arc length: The length of a curve from $t = a$ to $t = b$ is:

$$s = \int_a^b |\vec{v}(t)|\, dt = \int_a^b \sqrt{\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)^2 + \left(\frac{\mathrm{d}y}{\mathrm{d}t}\right)^2 + \left(\frac{\mathrm{d}z}{\mathrm{d}t}\right)^2}\, dt$$

In differential form:

$$ds = |\vec{v}|\, dt = \sqrt{dx^2 + dy^2 + dz^2}$$

Arc length parameter: We can reparametrize a curve using arc length s as the parameter. Then:

$$\frac{\mathrm{d}\vec{r}}{\mathrm{d}s} = \vec{T}$$

is automatically a unit tangent vector (since $\left|\frac{\mathrm{d}\vec{r}}{\mathrm{d}s}\right| = 1$).

Example: Consider the helix $\vec{r}(t) = \cos(t)\hat{\imath} + \sin(t)\hat{\jmath} + t\hat{k}$ for $0 \le t \le 2\pi$.

Velocity vector:

$$\vec{v}(t) = -\sin(t)\hat{\imath} + \cos(t)\hat{\jmath} + \hat{k}$$

Speed:

$$|\vec{v}(t)| = \sqrt{\sin^2(t) + \cos^2(t) + 1} = \sqrt{2}$$

Arc length from $t = 0$ to $t = 2\pi$:

$$s = \int_0^{2\pi} \sqrt{2}\, dt = 2\pi\sqrt{2}$$

Unit tangent vector:

$$\vec{T}(t) = \frac{1}{\sqrt{2}}\Big(-\sin(t)\hat{\imath} + \cos(t)\hat{\jmath} + \hat{k}\Big)$$

Connection to tensor calculus: In general curvilinear coordinates, the tangent vectors to coordinate curves form the basis vectors of the coordinate system. Understanding how tangent vectors transform is fundamental to tensor analysis.

2 Basis vectors as partial derivatives and Jacobian

In two-dimensional Euclidean space, a point is described by Cartesian coordinates

$$(x, y)$$

The position vector is

$$\mathbf{R} = x\vec{e_x} + y\vec{e_y}$$

where the basis vectors are

$$\vec{e_x} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad \vec{e_y} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

These satisfy:

$$\vec{e_x} \cdot \vec{e_x} = 1, \quad \vec{e_y} \cdot \vec{e_y} = 1, \quad \vec{e_x} \cdot \vec{e_y} = 0.$$

Thus, $\{\vec{e_x}, \vec{e_y}\}$ is an orthonormal basis.
A key feature:

$$\frac{\partial \vec{e_x}}{\partial x} = \frac{\partial \vec{e_x}}{\partial y} = \frac{\partial \vec{e_y}}{\partial x} = \frac{\partial \vec{e_y}}{\partial y} = 0.$$

The basis vectors are constant everywhere in space.

We now define polar coordinates (r, θ) by the transformation

$$x = r\cos\theta, \qquad y = r\sin\theta.$$

The position vector becomes

$$\mathbf{R} = r\cos\theta\,\vec{e_x} + r\sin\theta\,\vec{e_y}$$

We define the radial unit vector as

$$\tilde{\vec{e_r}} = \cos\theta\,\vec{e_x} + \sin\theta\,\vec{e_y}.$$

This points directly away from the origin.
And then define the angular unit vector as

$$\tilde{\vec{e_\theta}} = -\sin\theta\,\vec{e_x} + \cos\theta\,\vec{e_y}$$

This points in the direction of increasing θ.
These vectors satisfy

$$\tilde{\vec{e_r}} \cdot \tilde{\vec{e_r}} = 1, \quad \tilde{\vec{e_\theta}} \cdot \tilde{\vec{e_\theta}} = 1, \quad \tilde{\vec{e_r}} \cdot \tilde{\vec{e_\theta}} = 0$$

So they form an orthonormal basis at each point, but polar basis vectors depend on θ

Compute derivatives:

$$\frac{\partial \tilde{\vec{e_r}}}{\partial \theta} = -\sin\theta\, \vec{e_x} + \cos\theta\, \vec{e_y} = \tilde{\vec{e_\theta}}$$

$$\frac{\partial \tilde{\vec{e_\theta}}}{\partial \theta} = -\cos\theta\, \vec{e_x} - \sin\theta\, \vec{e_y} = -\tilde{\vec{e_r}}$$

Thus,

$$\frac{\partial \tilde{\vec{e_r}}}{\partial r} = 0, \qquad \frac{\partial \tilde{\vec{e_\theta}}}{\partial r} = 0$$

Key result:

$$\frac{\partial \tilde{\vec{e_r}}}{\partial \theta} = \tilde{\vec{e_\theta}}, \qquad \frac{\partial \tilde{\vec{e_\theta}}}{\partial \theta} = -\tilde{\vec{e_r}}$$

The basis vectors rotate as θ changes.

Now if you think about this, to find out the directions of the basis vectors, what we do is following the directions of the coordinate lines and in the case of polar coordinates, for the $\tilde{\vec{e_\theta}}$ basis vector, since the coordinate lines are actually curved, we need to pick the straightest possible direction when moving along it, which results in being the vector tangent to the curve.

This reasoning gives us a hint for the next important reinterpretation of basis vectors, as represented by a partial derivative of the position vector R

$$\vec{e_x} \equiv \frac{\partial \vec{R}}{\partial x} \tag{1}$$

If you carefully think about it, it might look weird at first, but it will eventually make sense. Think about **R** as a function of the coordinates x and y for example, and then consider adding a small value Δx to the x coordinate. The difference between these two position vectors results in a vector that is parallel to the x coordinate axis. So this resultant vector has the direction of the basis vector $\vec{e_x}$.
This observation is interesting, but we need to solve tha fact that the magnitude of this vector is varying , depending on the value of the Δx used.
This can be solved by dividing the resultant vector by the quantity Δx. By doing so, we obtain a vector pointing in the x-basis vector direction and magnitude 1...now we can proceed further by taking the limit as Δx approaches to 0:

$$\frac{\partial \vec{R}}{\partial x} = \lim_{\Delta x \to 0} \frac{\vec{R}(x + \Delta x, y) - \vec{R}(x, y)}{\Delta x} \tag{2}$$

Since this derivative always has a length of one and points the x-axis, we can regard it as the basis vector:

$$\vec{e_x} \equiv \frac{\partial \vec{R}}{\partial x} \tag{3}$$

This reasoning can be easily applied to the other direction, but generalized further to any coordinate system.
The neat thing about this construction is that you can do it for any position vector, and you will still get your basis vectors. Fundamentally, this trick work because, in obtaining a slightly off position vector, a small change in coordinate is added in one coordinate only while keeping other coordinates constant. Then, we divide the resultant vector (obtained from the difference between the original position vector and the new slightly off one) by that small change to normalize the vector.

Now you should remember when we discussed, in the other notes for "Tensors for Beginners" about the forward and backward transformations when we move from one basis vector coordinate system to another.
Turns out that those transformation rules can be obtained easily once we introduced this above concept about basis vectors being partial derivatives of the position vector with respect to the coordinate variables.
And also, that these transformation matrices have a proper name: **the Jacobian matrices**

Let's say we want to find the forward transformation matrix (the Jacobian) matric when moving from cartesian to polar coordinates. This would be just a matter of applying the chain rule:

$$\tilde{\vec{e_r}} = \frac{\partial \vec{R}}{\partial r} = \frac{\partial \vec{R}}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial \vec{R}}{\partial y}\frac{\partial y}{\partial r} = \frac{\partial x}{\partial r}\vec{e_x} + \frac{\partial y}{\partial r}\vec{e_y}$$
$$\tilde{\vec{e_\theta}} = \frac{\partial \vec{R}}{\partial \theta} = \frac{\partial \vec{R}}{\partial x}\frac{\partial x}{\partial \theta} + \frac{\partial \vec{R}}{\partial y}\frac{\partial y}{\partial \theta} = \frac{\partial x}{\partial \theta}\vec{e_x} + \frac{\partial y}{\partial \theta}\vec{e_y}$$

We know the transformation rules between cartesian and polar coordinates:

$$x = r\cos\theta, \qquad y = r\sin\theta.$$

so it's a matter of taking the derivatives to find that the Jacobian matrix, or forward matrix is:

$$\frac{\partial \vec{R}}{\partial r} = \frac{\partial \vec{R}}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial \vec{R}}{\partial y}\frac{\partial y}{\partial r} = cos\theta\frac{\partial \vec{R}}{\partial x} + sin\theta\frac{\partial \vec{R}}{\partial y}$$
$$\frac{\partial \vec{R}}{\partial \theta} = \frac{\partial \vec{R}}{\partial x}\frac{\partial x}{\partial \theta} + \frac{\partial \vec{R}}{\partial y}\frac{\partial y}{\partial \theta} = -r\sin\theta\frac{\partial \vec{R}}{\partial x} + r\cos\theta\frac{\partial \vec{R}}{\partial y}$$

So:

$$F = \begin{bmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{bmatrix} = \begin{bmatrix} cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{bmatrix} \tag{4}$$

And you can tackle, with the same derivation logic, the inverse Jacobian matrix, of backward transformation matrix, to obviously find out that the multiplication of the two yields the identity matrix.

Now, following a more tensorial approach, we can also re-write these transformation rules as:

$$\frac{\partial \vec{R}}{\partial p^i} = \frac{\partial \vec{R}}{\partial c^j}\frac{\partial c^j}{\partial p^i}$$
$$\frac{\partial \vec{R}}{\partial c^i} = \frac{\partial \vec{R}}{\partial p^j}\frac{\partial p^j}{\partial c^i}$$

3 Vectors as derivatives and their transformation rules

Allright we saw that basis vectors can be re-interpreted as partial derivative of the position vector along the coordinate axis.
We can sort of extend this logic to the vectors themselves. We saw that figuring out vector components in a particular basis is kind of easy to do for individual vectors, but now we're going do it for vector fields (i.e. a vector for every point in space), **but for now, we're going to deal with a slightly different type of vector field, and that's a vector field along a curve, so basically we're defining a vector at every point along a curve.**

The way we imagine vector fields along a curve is considering the set of tangent vectors to the curves. This gives us a vector field defined at every point on the curve.

A curve can be tought as a function that takes some input parameter λ and outputs the position vector $\mathbf{R}$ that traces our curve in the space.
If we want to have a vector tangent to this curve, we just need to consider taking the derivative of the position vector with respect to its parameter:

$$\frac{\mathrm{d}\vec{R}}{\mathrm{d}\lambda} \tag{5}$$

Now you may think that finding out the components of this vector in a specific basis can be difficult but it is really not, if we make use of the multivariable chain rule and expand it out in the coodinates we're working in.
For example, in the cartesian coordinates c^i and using the Einstein summation convention notation:

$$\frac{\mathrm{d}\vec{R}}{\mathrm{d}\lambda} = \frac{\mathrm{d}c^i}{\mathrm{d}\lambda}\frac{\partial\vec{R}}{\partial c^i} \tag{6}$$

At the end of the day, considering doing it in cartesian $\mathbb{R}^{\nvDash}$ for simplicity, these are equivalent (individual vector on the top, and vector field along a curve on the bottom):

$$\vec{v} = v^1\vec{e_1} + v^2\vec{e_2}$$
$$\frac{\mathrm{d}\vec{R}}{\mathrm{d}\lambda} = \frac{\mathrm{d}x}{\mathrm{d}\lambda}\frac{\partial\vec{R}}{\partial x} + \frac{\mathrm{d}y}{\mathrm{d}\lambda}\frac{\partial\vec{R}}{\partial y}$$

The same can be applied to whatever coordinate system we're working in, so to give you an example, let's say we have the circular curve of radius 2, centered at the origin in $\mathbb{R}^{\nvDash}$, defined by:

$$\vec{R}(\lambda) = (x(\lambda), y(\lambda)) = (2\cos\lambda, 2\sin\lambda) \tag{7}$$

The vector field along the curve is made up of the tangent vectors along the curve, that can be expanded as defined:

$$\begin{aligned}\frac{\mathrm{d}\vec{R}}{\mathrm{d}\lambda} &= \frac{\mathrm{d}x}{\mathrm{d}\lambda}\frac{\partial\vec{R}}{\partial x} + \frac{\mathrm{d}y}{\mathrm{d}\lambda}\frac{\partial\vec{R}}{\partial y}\\ &= -2\sin\lambda\frac{\partial\vec{R}}{\partial x} + 2\cos\lambda\frac{\partial\vec{R}}{\partial y}\end{aligned}$$

The components can also be written, for the cartesian basis, as a vector column:

$$\begin{bmatrix}-2\sin\lambda\\ 2\cos\lambda\end{bmatrix}_{\frac{\partial}{\partial c^i}} \tag{8}$$

We can do the same in polar coordinates r, θ, where the same curve is defined as:

$$\vec{R}(\lambda) = (r(\lambda), \theta(\lambda)) = (2, \lambda) \tag{9}$$

$$\begin{aligned}\frac{\mathrm{d}\vec{R}}{\mathrm{d}\lambda} &= \frac{\mathrm{d}r}{\mathrm{d}\lambda}\frac{\partial\vec{R}}{\partial r} + \frac{\mathrm{d}\theta}{\mathrm{d}\lambda}\frac{\partial\vec{R}}{\partial\theta}\\ &= 0\frac{\partial\vec{R}}{\partial r} + 1\frac{\partial\vec{R}}{\partial\theta}\end{aligned}$$

So:

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix}_{\frac{\partial}{\partial p^i}} \tag{10}$$

Now, as you can already imagine, like we saw that for individual vectors, the components are contravariant (i.e. they transform the opposite way of the basis vectors), we're going to simply show that this is valid indeed in this vector fields case.

We already saw in the previous section, the transformation rules of the basis vectors using the Jacobain and Inverse Jacobian:

$$\boxed{\begin{aligned} \frac{\partial \vec{R}}{\partial p^i} &= \frac{\partial \vec{R}}{\partial c^j}\frac{\partial c^j}{\partial p^i} && \text{Forward Transform} \\ \frac{\partial \vec{R}}{\partial c^i} &= \frac{\partial \vec{R}}{\partial p^j}\frac{\partial p^j}{\partial c^i} && \text{Backward Transform} \end{aligned}} \tag{11a, 11b}$$

Now we can just use these and replace the basis vectors with their new basis transformation rule:

$$\frac{\mathrm{d}\vec{R}}{\mathrm{d}\lambda} = \frac{\mathrm{d}c^i}{\mathrm{d}\lambda}\frac{\partial \vec{R}}{\partial c^i} = \frac{\mathrm{d}c^i}{\mathrm{d}\lambda}\left(\frac{\partial \vec{R}}{\partial p^j}\frac{\partial p^j}{\partial c^i}\right) = \left(\frac{\mathrm{d}c^i}{\mathrm{d}\lambda}\frac{\partial p^j}{\partial c^i}\right)\frac{\partial \vec{R}}{\partial p^j} \tag{12}$$

This is a linear combination of the new basis vector p^j, and since we know that we can write:

$$\frac{\mathrm{d}\vec{R}}{\mathrm{d}\lambda} = \frac{\mathrm{d}p^i}{\mathrm{d}\lambda}\frac{\partial \vec{R}}{\partial p^i} \tag{13}$$

This means that:

$$\frac{\mathrm{d}p^i}{\mathrm{d}\lambda} = \frac{\mathrm{d}c^j}{\mathrm{d}\lambda}\frac{\partial p^i}{\partial c^j} \tag{14}$$

Which shows that the vector field components transform using the backward transformation (inverse Jacobian)

We can actually test this out in the previous example right? So we have the components being, in both coordinate systems:

$$\begin{bmatrix} -2\sin\lambda \\ 2\cos\lambda \end{bmatrix}_{\frac{\partial}{\partial c^i}} \qquad \begin{bmatrix} 0 \\ 1 \end{bmatrix}_{\frac{\partial}{\partial p^i}}$$

And we also know the Inverse Jacobian in the case of polar-to-cartesian coordinates, being:

$$B = J^{-1} = \begin{bmatrix} x/r & y/r \\ -y/r^2 & x/r^2 \end{bmatrix} = \begin{bmatrix} \cos\lambda & \sin\lambda \\ -1/2\sin\lambda & 1/2\cos\lambda \end{bmatrix} \tag{15}$$

Let's check that:

$$
\begin{aligned}
B\begin{bmatrix}-2\sin\lambda\\ 2\cos\lambda\end{bmatrix}_{\frac{\partial}{\partial c^i}} &= \begin{bmatrix}\cos\lambda & \sin\lambda\\ -1/2\sin\lambda & 1/2\cos\lambda\end{bmatrix}\begin{bmatrix}-2\sin\lambda\\ 2\cos\lambda\end{bmatrix}_{\frac{\partial}{\partial c^i}}\\
&= \begin{bmatrix}-\cos\lambda 2\sin\lambda + \sin\lambda 2\cos\lambda\\ -1/2\sin\lambda(-2\sin\lambda) + 1/2\cos\lambda 2\cos\lambda\end{bmatrix}_{\frac{\partial}{\partial p^i}}\\
&= \begin{bmatrix}0\\ 1\end{bmatrix}_{\frac{\partial}{\partial p^i}}
\end{aligned}
$$

Same can be checked on the opposite direction applying the forward/jacobian transformation.

One last thing to mention before terminating this section is in terms of notation: you notice that the position vector appears in a lot of them, so what we're going to do from here onward, is get rid of the position vector symbol.
This might seem like a weird choice of notation, but from now on, for example, the $\mathbb{R}^2$ cartesian basis vectors will be written as:

$$
\begin{aligned}
\vec{e_x} &= \frac{\partial}{\partial x}\\
\vec{e_y} &= \frac{\partial}{\partial y}
\end{aligned}
$$

where basically we consider basis vectors these partial derivative operators.
If you think about this, it's not very weird at the end, as derivative operators are linear, as they can be scaled and added like vectors do, so they have both a magnitude and a direction. They also obey the forward and backward transformation rules that we would expect vectors to obey.

The main logical reason to get rid of the position vector symbol is that, the position vector definition requires an origin to be set, and later on when we're going to deal with curved surfaces and manifolds, it's not easy to define an origin point.
So, using a position vector without an origin does not make a lot of sense, but we're still going to be able to use vectors and vector fields as long as we have these derivative operators to work with.

The whole main idea is the following:
Basis transforms with the Jacobian;
Components transform with the inverse Jacobian;
Their product gives the identity;
Therefore the geometric vector is coordinate-invariant.

$$\frac{\mathrm{d}}{\mathrm{d}\lambda} = \frac{\mathrm{d}p^i}{\mathrm{d}\lambda}\frac{\partial}{\partial p^i} = \frac{\mathrm{d}c^i}{\mathrm{d}\lambda}\frac{\partial}{\partial c^i} \qquad \text{Total derivative along a curve (chain rule)} \tag{16a}$$

$$\frac{\partial}{\partial p^i} = \frac{\partial}{\partial c^j}\frac{\partial c^j}{\partial p^i} \qquad \text{Coordinate basis transformation} \tag{16b}$$

$$\frac{\partial}{\partial c^i} = \frac{\partial}{\partial p^j}\frac{\partial p^j}{\partial c^i} \qquad \text{Inverse coordinate basis transformation} \tag{16c}$$

$$\frac{\mathrm{d}p^i}{\mathrm{d}\lambda} = \frac{\mathrm{d}c^j}{\mathrm{d}\lambda}\frac{\partial p^i}{\partial c^j} \qquad \text{Contravariant component transformation} \tag{16d}$$

$$\frac{\mathrm{d}c^i}{\mathrm{d}\lambda} = \frac{\mathrm{d}p^j}{\mathrm{d}\lambda}\frac{\partial c^i}{\partial p^j} \qquad \text{Inverse component transformation} \tag{16e}$$

3.1 Hold on - Why are derivative operators vectors?

Let's here make some clarity about what we've discussed so far in the previous two sections. We brought up this new way of thinking about basis vectors, where we went from considering basis vectors as derivative of the position vector wrt the coordinate variables, to them being defined by the partial derivative operators themselves.
Let's start by saying that this way of thinking will be appreciated more when we'll start thinking anout curved (differential) geometry, but we chose to introduce it right now, even though we're still working in flat space, to already start using these ideas and notations beforehand.

We just a bit ahead of ourselves here, to explain the main reason of already working with this, so you get an idea of what we're going to deal with, and then we'll get back to our flat space for now.

We saw that if we consider the difference between two position vectors, a small distance h apart from each other in the direction of one coordinate axis, divide this by this distance h, and consider the limit when h approaches 0, gives us as result the basis vector of magnitude 1 in the considered axis direction. We we said that this position vector derivative along the axis-variable and the unit vector along this axis are equivalent.

We also shown that, for tangent vectors along a curve, parametrized by the paramter λ, considering the position vector at a certain position λ and another one a small step h away, we can compute the tangent vector to this curve at position λ, by taking the difference and consider the limit when h approaches 0.

And we can expand the tangent vector out as a linear combination of the basis vectors using the chain rule. Now all of this, has been discussed working on the flap 2D plane.

Let's try to move further and consider this in a curved space, for example the surface of the earth, to see how the situation is different.

If we consider a point moving on the surface of the earth from one position to another, and we want to compute its velocity, we need to find the tangent vector to the curve he moves along. So what do we do? We can take the derivative of the position vector wrt to the parameter we use to define this curve.
Okay, so how do we define the position vector? We need to define an origin, right? We can think of putting this origin at the center of the sphere (earth)....but this is a bit of a different situation than before when working in the flat space?

Why? Well first because the origin does not live on the earth's 2D surface, and second because the velocity vector itself, also does not live on the earth's 2D surface, since it extends to the outside space.
They both are external to the surface we're studying, and indeed this is referred to "extrinsic geometry"

Now, this may not seem a problem, if we consider the 3D surface of a globe, from an external perspective of a 3D space.
But what if instead of this situation, we're forced to look at the surface of the earth on a map?

How do we define position vectors here? There's not really a convenient outside place where we can put the origin here, there's no external place for us to use. You might think that you can potentially use any place on the map to put the origin, choosing a random placefor it and drawing a straight line representing the position vector.

There's a problem with this approach though, vectors are supposed to be straight, but the path from the origin to the tip of the vector is not really straight, because the surface in reality is not flat.

The method of studying curved surfaces where we're not allowed to leave the surface is called intrinsic geometry.

When we'll study General relativity, we're going to deal with a 4-dimensional curved spacetime geometry, the situation we're running on is exactly this. Spacetime is curved so we cannot really run straight lines, hence we cannot draw position vectors.

And we cannot pick an origin point outside of this 4-dim space, because that would mean picking a point outside universe, and that's completely meaningless.

So the summary is that, we're forced to use intrinsic geometry in General Relativity, so how are we supposed to define things like velocity, acceleration etc if we're not allowed to use position vectors?

The solution is, to realize that, while we cannot draw straight lines on a curved surface, we can still draw a curved path, and on a map, we can use grid lines, so we can still define derivatives wrt these curves. So we can use these derivative operators to talk about different directions on the surface.
The partial derivative wrt to x for example, points to the x direction, the partial wrt to y points to the y direction and so on.

If we have some path traveling around in our curved space, we can still consider the direction that the path is pointing to, using the derivative wrt the curve parameter used, and we break this direction up in terms of our x, y components, using the multi-variable chain rule as we saw.

Formally, our new notation with basis vectors defined as partial derivative operators along the axis directions, form a vector space called the "Tangent Vector Space" T_pM, which is the vector space of derivatives at some point p on a surface M

4 Differentials as covectors and their transformation rules

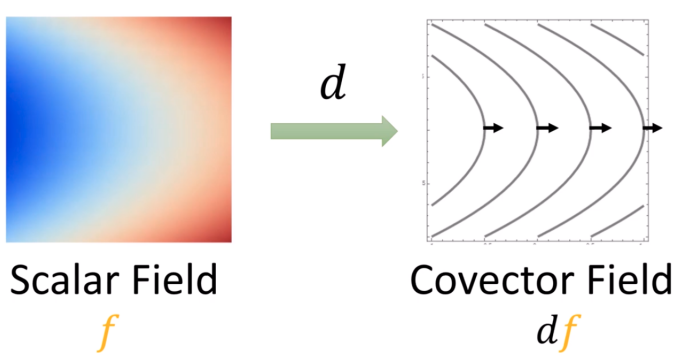
Let's start this chapter by reminind ourselves what differentials are and covectors are. We know that differentials in Integrals, like dx are small changes in the variable of integration, in this case it's a small change in the variable x.

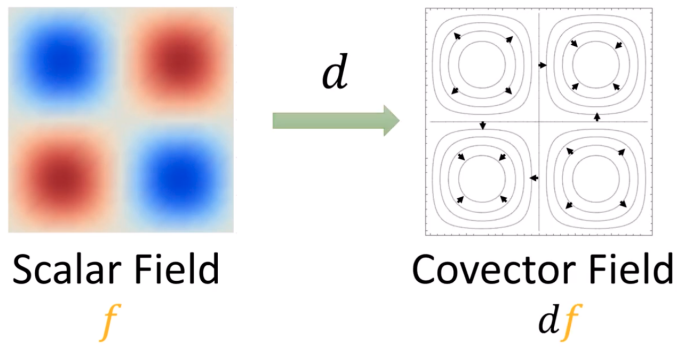
Covectors instead, are these functionals acting on vectors, and outputting scalars, which can be visualized as stacks of lines that indeed eat a vector and produce a scalar. The scalar can be tought as the number of times the vector pierces the stack of lines.

So like, how can these two concepts be linked together, and actually how the hell are they the same thing?

The key of understanding this, is the re-interpretation of the d symbol of the differentials. The old way of thinking about it, is that this symbol takes the variable and gives a small change of this variable.
The new interpretation is that d is an operator that takes a scalar field f and outputs a df covector field. A scalar field is a field where a scalar f is defined at every point in space, a covector field df is a field where a covector is defined at every point in space.

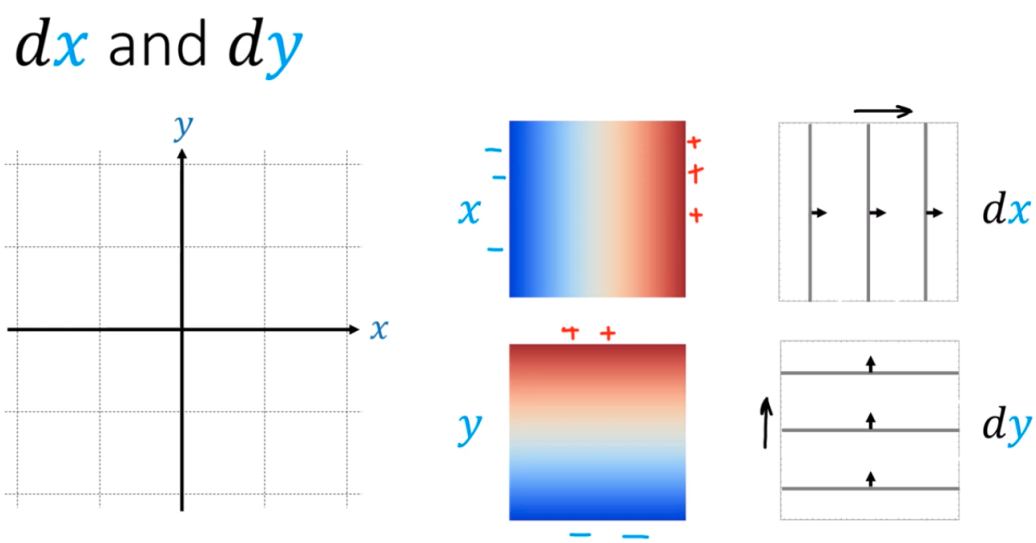
So how does this d operator work? A couple of pictures are worth a thousand words here:



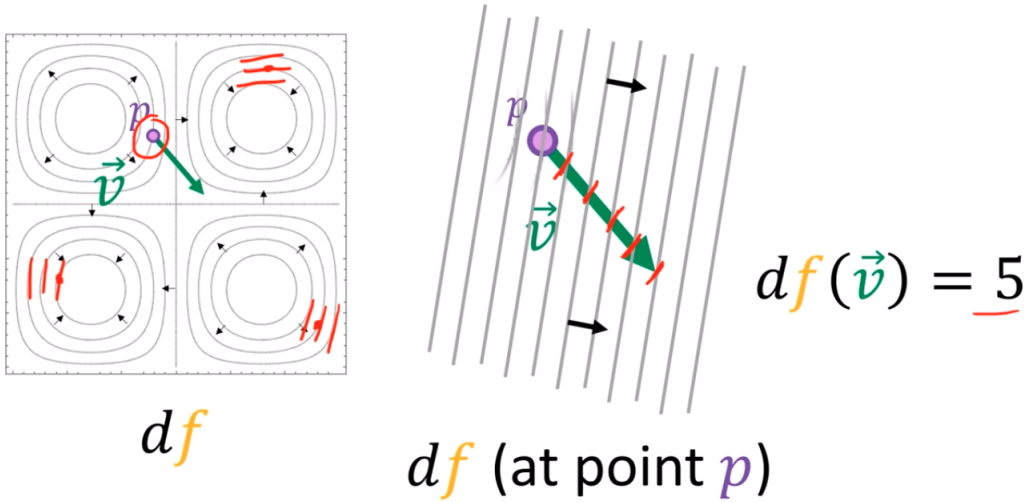


So like, the operator d takes a function and outputs the function's level sets. Sometimes, they're called 0-forms (functions) and 1-forms (level sets).

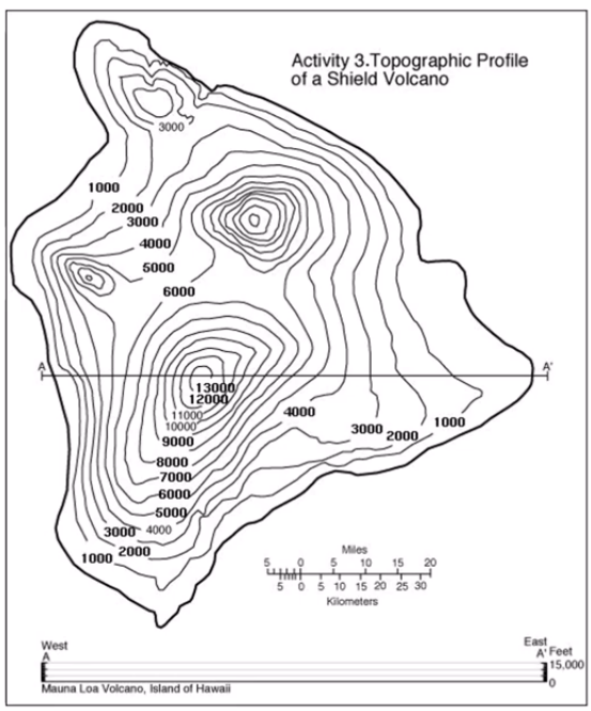
Let's give another example, showing the covector fields of the cartesian coordinates dx and dy:



Now, if this operator is really a covector field, we know that covectors eat vectors to output scalars, right? How does this work considering what we've seen so far?



And it's also quite straight forward to show that covector fields are linear. But whet's the geometrical meaning of $df(\vec{v})$



We can think of these level sets of a map as a covector field.
When acting on a vector, $df(\vec{v})$ is proportional to the steepness of f in the direction of $\vec{v}$ and also proportional to the length of $\vec{v}$.

So: $df(\vec{v})$ **tells us the rate of change of** f **when moving at velocity** $\vec{v}$

But then, this reminds us of the definition of directional derivative, isn't it? At the end of the day, $df(\vec{v})$ is the **directional derivative** of the function in the vector direction.

$$df(\vec{v}) = D_{\vec{v}}f = \frac{\partial f}{\partial \vec{v}} \tag{17}$$

4.1 Differential forms components and their transformation

Just as we can write vectors as linear combinations of special basis vectors, scaled by some components, we can break covector fields, or differential forms, as linear combination of some special differential forms basis scaled by their components.

$$df = Adx + Bdy \tag{18}$$

Let's start by checking how differential forms act on our special basis vectors $\frac{\partial}{\partial x}, \frac{\partial}{\partial y}$:

$$df(\vec{e_x}) = df\left(\frac{\partial}{\partial x}\right) = \frac{\partial f}{\partial x} \tag{19}$$

That's the directional derivative of f in the x direction.
Same for the other direction:

$$df(\vec{e_y}) = df\left(\frac{\partial}{\partial y}\right) = \frac{\partial f}{\partial y} \tag{20}$$

Now we can look at the simple scalar field x, and the scalar value at a point is just the value of x at that point, positives on the right, negative on the left.
The corresponding covector field is dx, represented by vertical level sets directed to the right.
So:

$$\begin{aligned} dx(\vec{e_x}) &= dx\left(\frac{\partial}{\partial x}\right) \\ &= \frac{\partial x}{\partial x} = 1 \\ dx(\vec{e_y}) &= dx\left(\frac{\partial}{\partial y}\right) \\ &= \frac{\partial x}{\partial y} = 0 \end{aligned}$$

Same process can be applied to the scalar field y, obtaining in summary:

$$dx\left(\frac{\partial}{\partial x}\right) = \frac{\partial x}{\partial x} = 1 \qquad dx\left(\frac{\partial}{\partial y}\right) = \frac{\partial x}{\partial y} = 0$$

$$dy\left(\frac{\partial}{\partial x}\right) = \frac{\partial y}{\partial x} = 0 \qquad dy\left(\frac{\partial}{\partial y}\right) = \frac{\partial y}{\partial y} = 1$$

Which can be re-written, considering $c^1 = x, c^2 = y$:

$$dc^i\left(\frac{\partial}{\partial c^j}\right) = \frac{\partial c^i}{\partial c^j} = \delta^i_j \tag{21}$$

As when we discussed the special dual basis (co)vectors in the "Tensor for Beginners" notes, these special differential forms dx, dy are the dual basis for differential forms.
To find the coefficients, let's just consider a generic function $f(x, y)$ and a curve parametrized by the parameter λ.
The tangent vectors along the curve are given by $\frac{\partial}{\partial\lambda}$.
Let's consider the rate of change of this function as we travel along the curve, which is given by the derivative $\frac{df}{d\lambda}$

Now remember this, the covector field df, acting on the vector $d/d\lambda$ is equal to $df/d\lambda$ so we can re-write:

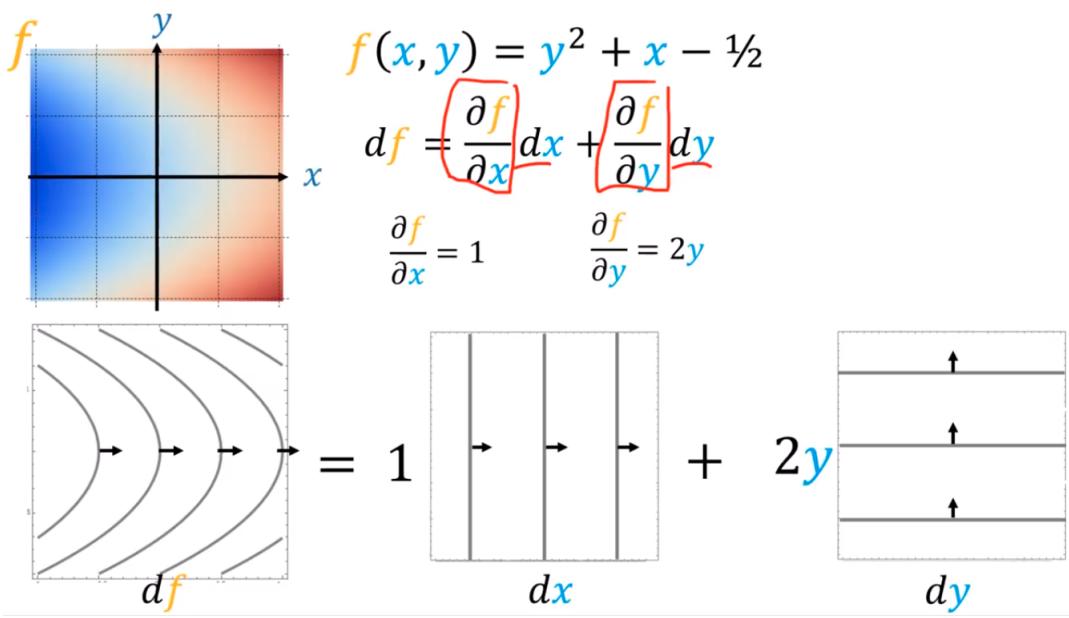
$$\begin{aligned}
\frac{df}{d\lambda} &= \frac{\partial f}{\partial x}\frac{dx}{d\lambda} + \frac{\partial f}{\partial y}\frac{dy}{d\lambda} \\
df\left(\frac{d}{d\lambda}\right) &= \frac{\partial f}{\partial x}dx\left(\frac{d}{d\lambda}\right) + \frac{\partial f}{\partial y}dy\left(\frac{d}{d\lambda}\right) \\
df\left(\frac{d}{d\lambda}\right) &= \left(\frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy\right)\left(\frac{d}{d\lambda}\right)
\end{aligned}$$

So from the last equation, we can deduce that these two quantities are the same vector field:

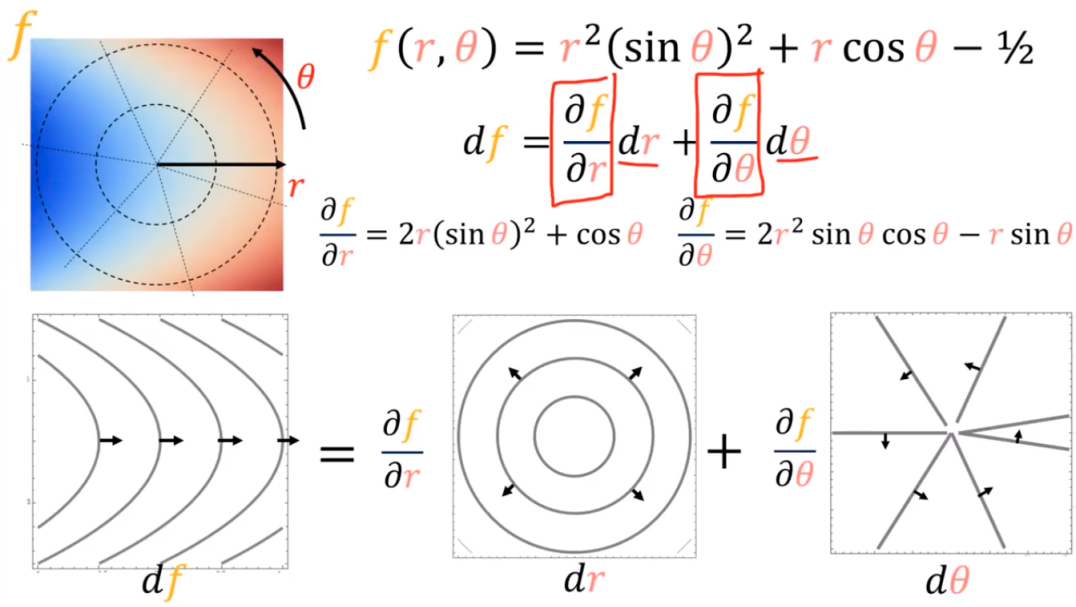
$$df = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy \tag{22}$$

And this is indeed what we were looking for, writing the covector field as linear combination of dx, dy, so dx, dy form a basis for the set of covector fields. They're the dual basis for differential forms.

Let's see a more concrete example to understand this better. Consider this scalar field, we can write down the covector field in terms of the basis vectors in this way:



The same can be done in other coordinate systems, like polar ones:



Now, how would both the covector basis field and the covector field components transform? Well we just showed that any covector field can be written as linear combination of basis covector field dx and dy.

But since this is valid for any covector field, we can expand our new polar covector field basis:

$$dr = \frac{\partial r}{\partial x}dx + \frac{\partial r}{\partial y}dy$$
$$d\theta = \frac{\partial \theta}{\partial x}dx + \frac{\partial \theta}{\partial y}dy$$

If you recall, those coefficients are just the entries of the Inverse Jacobian Matrix B or J^{-1}. What this means is that, covector basis field transform contravariantly.
In Einstein notation, and then in matrix notation, we can write:

$$dp^i = \frac{\partial p^i}{\partial c^j}dc^j$$
$$\begin{bmatrix} dr \\ d\theta \end{bmatrix} = \begin{bmatrix} \frac{\partial r}{\partial x} & \frac{\partial r}{\partial y} \\ \frac{\partial \theta}{\partial x} & \frac{\partial \theta}{\partial y} \end{bmatrix} \begin{bmatrix} dx \\ dy \end{bmatrix}$$

Remember that when writing in matrix notation, to make formulas work, we need to write contravariant components as column vectors.
The other way around is also possible, so we'd have that to go from the new to the hold covector basis field, the Jacobian, or forward matrix is used:

$$dc^i = \frac{\partial c^i}{\partial p^j}dp^j$$
$$\begin{bmatrix} dx \\ dy \end{bmatrix} = \begin{bmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{bmatrix} \begin{bmatrix} dr \\ d\theta \end{bmatrix}$$

Now what about the covector field components?
Quite straight forward just using the usual chain rule:

$$\frac{\partial f}{\partial p^j} = \frac{\partial c^i}{\partial p^j}\frac{\partial f}{\partial c^i}$$
$$\frac{\partial f}{\partial c^j} = \frac{\partial p^i}{\partial c^j}\frac{\partial f}{\partial p^i}$$

So while the basis covector fields obey the contravariant transformation rule, the covector field components obey the covariant one.
A full summary is depicted below:

$$\frac{d}{d\lambda} = \frac{dc^i}{d\lambda}\frac{\partial}{\partial c^i} = \frac{dp^i}{d\lambda}\frac{\partial}{\partial p^i}$$

$$df = \frac{\partial f}{\partial c^i}\boldsymbol{dc^i} = \frac{\partial f}{\partial p^i}\boldsymbol{dp^i}$$

Basis Vectors

Covariant

$$\frac{\partial}{\partial p^j} = \frac{\partial c^i}{\partial p^j}\frac{\partial}{\partial c^i}$$
$$\frac{\partial}{\partial c^j} = \frac{\partial p^i}{\partial c^j}\frac{\partial}{\partial p^i}$$

Basis Covectors

Contravariant

$$\boldsymbol{dp^i} = \frac{\partial p^i}{\partial c^j}\boldsymbol{dc^j}$$
$$\boldsymbol{dc^i} = \frac{\partial c^i}{\partial p^j}\boldsymbol{dp^j}$$

Vector Components

Contravariant

$$\frac{dp^i}{d\lambda} = \frac{\partial p^i}{\partial c^j}\frac{dc^j}{d\lambda}$$
$$\frac{dc^i}{d\lambda} = \frac{\partial c^i}{\partial p^j}\frac{dp^j}{d\lambda}$$

Covector Components

Covariant

$$\frac{\partial f}{\partial p^j} = \frac{\partial c^i}{\partial p^j}\frac{\partial f}{\partial c^i}$$
$$\frac{\partial f}{\partial c^j} = \frac{\partial p^i}{\partial c^j}\frac{\partial f}{\partial p^i}$$

5 Integration with differential forms

What we've discussed so far has shown that out operator d takes a scalar field f and produces a covector field df which is the level sets of f oriented towards the positive values of f.
And this covector field df can behave like a function that acts on vectors.
If applied to a vector $\vec{v}$, we think about zooming on the covector at the point of $\vec{v}$ and count how many times the vector piereces to get the output.

So this reinterpretation of the operator d gives us covector fields which are also called differential forms. How does this relate back to the differential we see in integrals, which represents a small change in the varibale, for example dx?

The short answer is that, every single integral involves:

- a differential form (covector field)
- a path

$$\int_a^b f(x)dx \tag{23}$$

The result of the integral is just the number of covector stacks pierced by the path.

To prove this, we're going to explore a couple of examples from Physics, but first let's quickly review two important theorems from calculus:

- Fundamental Theorem of Calculus

$$\int_a^b f(x)dx = F(b) - F(a)$$
$$\int_a^b \frac{dF}{dx}dx = F(b) - F(a)$$

- Gradient Theorem

$$\int_{P[a,b]} \boldsymbol{\nabla} F \cdot dr = F(b) - F(a)$$

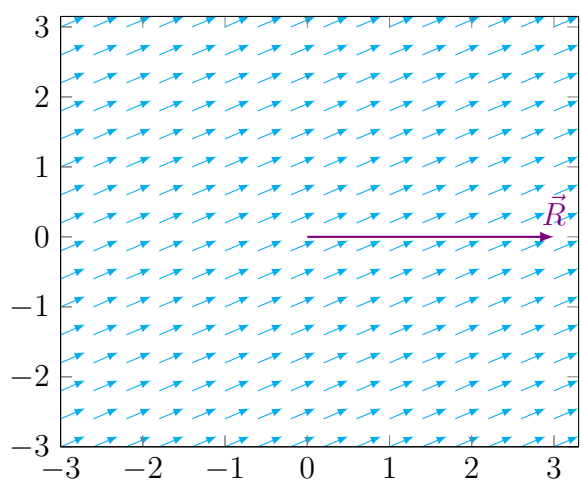
Both of them involve integrals over a path from a to b, where the integral itself only depends on the values at the endpoints.
We're going to prove the Gradient theorem to show why differential forms are strictly tied to integrals.

Recall what work is in physics, it's the application of a force to an object along some distance, the force dot product with the displacement vector:

$$W = \vec{F} \cdot \vec{R} \tag{24}$$

The dot product actually tells us how much of the force was used to move the object along the displacement vector.
Let's say that we have some force vector defined as:

$$\vec{F} = 2\vec{e_1} + 1\vec{e_2} \tag{25}$$



And our vector giving us the direction is:

$$\vec{R} = 3\vec{e_1} + 0\vec{e_2} \tag{26}$$

The total work would be $W = \vec{F} \cdot \vec{R} = 6$ Joules

Now, for a bit more meaningful example, consider the force field due to the gravitational field that the earch exercit on a smaller mass m, at a distance $\vec{R}$:

$$\vec{F} = \frac{GMm}{\|\vec{R}^2\|}(-\vec{e_r}) \tag{27}$$

Giving a curved path with an object traveling along, how do we compute the work done on this object, by the gravitational field. The formula we used before is not valid here, as the force field is different at every point in space, and the path we're traveling along is curved.

We actually need to perform an integration along the path like:

$$W = \int_P \vec{F}_G \cdot d\vec{R} \tag{28}$$

This $d\vec{R}$, what it really is, it's just a short way of seeing $\frac{d\vec{R}}{d\lambda}d\lambda$.
Where λ is like the time parameter that increases as we move along the curve. So really, $\frac{d\vec{R}}{d\lambda}$ is the velocity vector or the tangent vector along the path.

$$W = \int_P \vec{F}_G \cdot \frac{d\vec{R}}{d\lambda}d\lambda \tag{29}$$

Let's compute this for a couple of different paths, circular and straight out from earth:

Circular path $\vec{R}(\lambda) = (r = 2, \theta = \lambda)$

$$\begin{aligned}
\frac{d\vec{R}}{d\lambda} &= \frac{dr}{d\lambda}\frac{d\vec{R}}{dr} + \frac{d\theta}{d\lambda}\frac{d\vec{R}}{d\theta} \\
\frac{d\vec{R}}{d\lambda} &= 0\vec{e_r} + 1\vec{e_\theta} \\
W &= \int_P \left(\frac{GMm}{\|\vec{R}^2\|}(-\vec{e_r})\right) \cdot (\vec{e_\theta})d\lambda = 0
\end{aligned}$$

Because the two basis vectors are orthogonal to each other.
This means that over a circular path with constant radius, no work gets done by the gravitational field, as there's no component of its force along the curve.

Straight path ourwards from earth $\vec{R}(\lambda) = (r = \lambda, \theta = 0)$

$$
\begin{aligned}
\frac{d\vec{R}}{d\lambda} &= 1\vec{e_r} + 0\vec{e_\theta} \\
W &= \int_P \left(\frac{GMm}{\|\vec{R}^2\|}(-\vec{e_r}) \right) \cdot (\vec{e_r}) d\lambda \\
&= GMm \left[\tfrac{1}{3}x^3 \right]_a^b \\
&= GMm \left(\tfrac{1}{b} - \tfrac{1}{a} \right)
\end{aligned}
$$

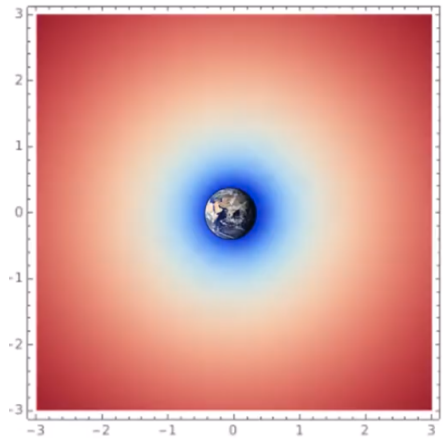
If we stop a moment to think, and look at the geometric picture of it, it's kind of not very intuitive that, whatever path we take, the integral always depend only on the endpoints (i.e. start and finish) of the path, even though we're actually considering all those tangent vectors along the path, and computing the dot product with the gravitational field.
It's pretty amazing, if not magical, that all those add up so that the total amount is only dependent on the endpoints, whatever strange curve we go along between points a and b, isnt' it?

There's indeed a better way to visualize this, that also makes this fact, that the work only depends on the start and end of the path, very intuitive and also kind of obvious.

To talk about this, we need to introduce the gravitational potential.
First we introduce the gravitational field, which is nothing more than the gravitational force we saw before, but that only depends on the source mass generating it, so without the small mass m.
This vector field can be written as the gradient of a scalar field, that we call gravitational potential:

$$
\begin{aligned}
\vec{F}_G &= m\vec{G} \\
\vec{G} &= -\boldsymbol{\nabla}\Phi \\
\vec{F}_G &= m(-\boldsymbol{\nabla}\Phi)
\end{aligned}
$$

The gravitational potential can be visualized as:



Now using this definition of the gravitational force as dependning on the gradient of the gravitational potential, we can write, considering cartesian coordinates:

$$
\begin{aligned}
W &= \int_P \vec{F_G} \cdot \frac{d\vec{R}}{d\lambda} d\lambda \\
&= \int_P m(-\boldsymbol{\nabla}\Phi) \cdot \frac{d\vec{R}}{d\lambda} d\lambda \\
\boldsymbol{\nabla}\Phi &= \frac{\partial \Phi}{\partial x}\vec{e_x} + \frac{\partial \Phi}{\partial y}\vec{e_y} \\
\frac{d\vec{R}}{d\lambda} &= \frac{dx}{d\lambda}\frac{d\vec{R}}{dx} + \frac{dy}{d\lambda}\frac{d\vec{R}}{dy} \\
&= \frac{dx}{d\lambda}\vec{e_x} + \frac{dy}{d\lambda}\vec{e_y}
\end{aligned}
$$

$$
\begin{aligned}
&\boldsymbol{\nabla}\Phi \cdot \frac{d\vec{R}}{d\lambda} \\
&\quad = \frac{\partial \Phi}{\partial x}\frac{dx}{d\lambda} + \frac{\partial \Phi}{\partial y}\frac{dy}{d\lambda}
\end{aligned}
$$

So that:

$$
\begin{aligned}
W &= \int_P m(-\boldsymbol{\nabla}\Phi) \cdot \frac{d\vec{R}}{d\lambda} d\lambda \\
&= -m \int_P \frac{d\Phi}{d\lambda} d\lambda \\
&= -m \int_P d\Phi \\
&= -m \left[\Phi(P_{end}) - \Phi(P_{start})\right]
\end{aligned}
$$

So really, when we write down the integral with the dot product between the gradient and the tangent vectors along the curve, we're sort of overcomplicating things, because all

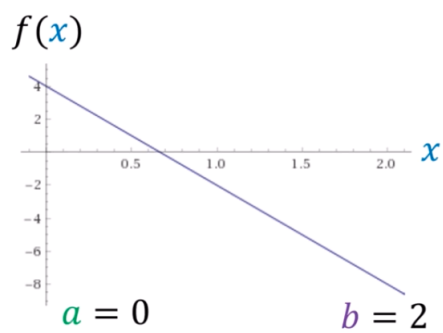
of it is just the expaneded version of $d\Phi$.
We just have to evaluate Φ at the end and at the start of the curve and then subtract.

Looking at the scalar field and not the associated vector field (given by the gradient of the scalar field) does indeed make more sense to compute the work.
Then, yet another way of looking at things, instead of looking at the scalar field Φ, we can look at the level sets of Φ, and when we draw a path through these level sets, if we can to compute the work along, we just look at where the path starts, where it ends, and see how much the function has changed. Basically by counting how many level sets, the path piereces from start to end.

We can move ahead with another of example, with a simple single variable integration.

$$\int_0^2 (-6x+4)dx$$

Integration is over the single variable x. If we represent this function and compute the integral as we are usually doing:



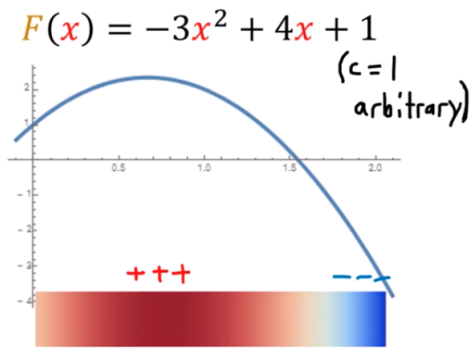
$$\begin{aligned}
&\int_0^2 (-6x+4)dx \\
&= \left[-3x^2+4x\right]_0^2 \\
&= \left[-3(2)^2+4(2)\right] - \left[-3(0)^2+4(0)\right] \\
&= -12+8 \\
&= -4
\end{aligned}$$

We can think of re-interpreting this using our new covector field vision:

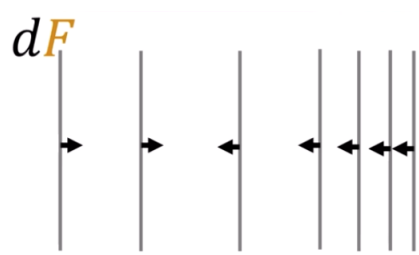
$$
\begin{aligned}
&\int_0^2 (-6x+4)dx \\
&= \int_0^2 \frac{dF}{dx} dx \\
&= \int_0^2 dF
\end{aligned}
$$

$$
\begin{aligned}
&\frac{dF}{dx} = -6x + 4 \\
&F(x) = -3x^2 + 4x + C
\end{aligned}
$$

Visualising it, it would look like this function here, but we can also visualize it as scalar field:



The associated covector field dF would look like this:



If you now plot the path from $x = 0$ to $x = 2$, you can clearly see yourself that, adding a $+1$ when the path pierces the covector field along its direction, and a -1 when on the opposite direction, the path crosses the covector field exactly $+1 - 1 - 1 - 1 - 1 - 1 = -4$ times, which is exactly the same value we got before using standard calculus.

6 The metric tensor and arch length - flat and curved spaces

We already saw in the *Tensor for beginners* notes, that the metric tensor is used to compute the length of a vector, the dot product between two vectors and the angle in between, and we also saw that it transforms using 2 covariant rules, hence is a (0,2)-tensor (bilinear form).

$$g : V \times V \to \mathbb{R}$$
$$g\left(\vec{v}, \vec{w}\right) \mapsto v^i w^j g_{ij}$$

$$\tilde{g_{ij}} = F_i^k F_j^l g_{kl}$$
$$g_{kl} = B_k^i B_l^j \tilde{g_{ij}}$$

We'll here focus on using the metric tensor to compute lengths of curves in flat space.

6.1 Arch length in flat spaces

We recall that in order to get the length of a curve in multi-variable calculus, basically we approximate the curve by a series of straight line segments, sum them up and take the limit as the Δ parameter used to parametrize the curve approaches to 0.
At the end of the day, the key to get the curve length is getting the magnitude of the vector $\frac{dR}{d\lambda}$ if λ is the parameter, and R the position vector used to describe a point on the curve. Which is nothing but the tangent velocity vector along the curve:

$$\text{arc length} = \int \left\| \frac{d\vec{R}}{d\lambda} \right\| d\lambda$$

Now, to get this magnitude, we can just compute the dot product of this vector with itself, as we know, and do that in cartesian coordinates for example:

$$\begin{aligned}
\left\| \frac{d\vec{R}}{d\lambda} \right\|^2 &= \frac{d\vec{R}}{d\lambda} \cdot \frac{d\vec{R}}{d\lambda} \\
&= \left(\frac{\partial \vec{R}}{\partial x} \frac{dx}{d\lambda} + \frac{\partial \vec{R}}{\partial y} \frac{dy}{d\lambda} \right) \cdot \left(\frac{\partial \vec{R}}{\partial x} \frac{dx}{d\lambda} + \frac{\partial \vec{R}}{\partial y} \frac{dy}{d\lambda} \right) \\
&= \left(\frac{dx}{d\lambda} \right)^2 \left(\frac{\partial \vec{R}}{\partial x} \cdot \frac{\partial \vec{R}}{\partial x} \right) + 2 \frac{dx}{d\lambda} \frac{dy}{d\lambda} \left(\frac{\partial \vec{R}}{\partial x} \cdot \frac{\partial \vec{R}}{\partial y} \right) + \left(\frac{dy}{d\lambda} \right)^2 \left(\frac{\partial \vec{R}}{\partial y} \cdot \frac{\partial \vec{R}}{\partial y} \right)
\end{aligned}$$

This ugly long product chain will look similar in other coordinate systems, like polar:

$$= \left(\frac{dr}{d\lambda} \right)^2 \left(\frac{\partial \vec{R}}{\partial r} \cdot \frac{\partial \vec{R}}{\partial r} \right) + 2 \frac{dr}{d\lambda} \frac{d\theta}{d\lambda} \left(\frac{\partial \vec{R}}{\partial r} \cdot \frac{\partial \vec{R}}{\partial \theta} \right) + \left(\frac{d\theta}{d\lambda} \right)^2 \left(\frac{\partial \vec{R}}{\partial \theta} \cdot \frac{\partial \vec{R}}{\partial \theta} \right)$$

If you carefully look at this, you will find out that it's very similar to when we computed the magnitude of a vector in our *Tensor for beginners* notes:

$$
\begin{aligned}
\|\vec{v}\|^2 &= \vec{v} \cdot \vec{v} \\
&= \left(v^1 \vec{e_1} + v^2 \vec{e_2}\right) \cdot \left(v^1 \vec{e_1} + v^2 \vec{e_2}\right) \\
&= \left(v^1\right)^2 \left(\vec{e_1} \cdot \vec{e_1}\right) + 2 v^1 v^2 \left(\vec{e_1} \cdot \vec{e_2}\right) + \left(v^2\right)^2 \left(\vec{e_2} \cdot \vec{e_2}\right) \\
&= \left(\tilde{v^1}\right)^2 \left(\tilde{\vec{e_1}} \cdot \tilde{\vec{e_1}}\right) + 2 \tilde{v^1} \tilde{v^2} \left(\tilde{\vec{e_1}} \cdot \tilde{\vec{e_2}}\right) + \left(\tilde{v^2}\right)^2 \left(\tilde{\vec{e_2}} \cdot \tilde{\vec{e_2}}\right)
\end{aligned}
$$

The key is indeed getting those dot products like we did in those initial notes, and those are nothing more than the components of the metric tensor in different coordinate systems.

$$
g_{c^i} = \begin{bmatrix} \frac{\partial \vec{R}}{\partial x} \cdot \frac{\partial \vec{R}}{\partial x} & \frac{\partial \vec{R}}{\partial y} \cdot \frac{\partial \vec{R}}{\partial x} \\ \frac{\partial \vec{R}}{\partial x} \cdot \frac{\partial \vec{R}}{\partial y} & \frac{\partial \vec{R}}{\partial y} \cdot \frac{\partial \vec{R}}{\partial y} \end{bmatrix} \qquad g_{p^i} = \begin{bmatrix} \frac{\partial \vec{R}}{\partial r} \cdot \frac{\partial \vec{R}}{\partial r} & \frac{\partial \vec{R}}{\partial \theta} \cdot \frac{\partial \vec{R}}{\partial r} \\ \frac{\partial \vec{R}}{\partial r} \cdot \frac{\partial \vec{R}}{\partial \theta} & \frac{\partial \vec{R}}{\partial \theta} \cdot \frac{\partial \vec{R}}{\partial \theta} \end{bmatrix}
$$

Allright, now using the Einstein notation to save some writing:

$$
\begin{aligned}
\left\| \frac{d\vec{R}}{d\lambda} \right\|^2 &= \frac{d\vec{R}}{d\lambda} \cdot \frac{d\vec{R}}{d\lambda} \\
&= \frac{dc^i}{d\lambda} \frac{dc^j}{d\lambda} \left(\frac{\partial \vec{R}}{\partial c^i} \cdot \frac{\partial \vec{R}}{\partial c^j} \right) = \frac{dc^i}{d\lambda} \frac{dc^j}{d\lambda} g_{ij} \\
&= \frac{dp^i}{d\lambda} \frac{dp^j}{d\lambda} \left(\frac{\partial \vec{R}}{\partial p^i} \cdot \frac{\partial \vec{R}}{\partial p^j} \right) = \frac{dp^i}{d\lambda} \frac{dp^j}{d\lambda} \tilde{g_{ij}}
\end{aligned}
$$

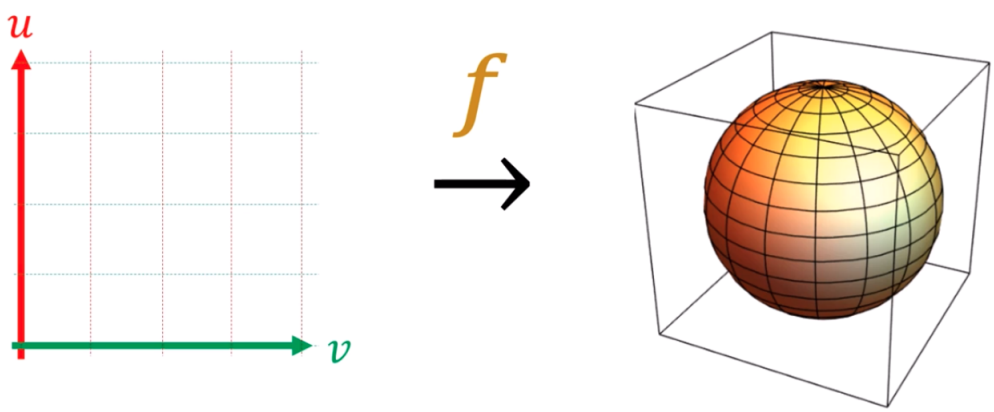
And its transformation rules:

$$
\begin{aligned}
\tilde{g_{ij}} &= \frac{\partial c^k}{\partial p^i} \frac{\partial c^l}{\partial p^j} g_{kl} \\
g_{kl} &= \frac{\partial p^i}{\partial c^k} \frac{\partial p^j}{\partial c^l} \tilde{g_{ij}}
\end{aligned}
$$

6.2 Arch length in curved spaces

Now we'll move on computing arc lengths for curved space, and in particular we're going to focus on the surface of sphere for our maine example.

The first question we can ask ourselves is, how do we describe a 2D surface mathematically? Well, the most common and straight forward way is to start with a flat 2D plane, with u and v coordinates, and then put this into some sort of function f which will stretch and transform the plane so that it becomes a 2D surface living in the 3D space basically:



The variables u and v become parameters of this vector function f (i.e. its **domain** D contained in $\mathbb{R}^2$). The range of the surfaces is contained in the three dimensional space $\mathbb{R}^3$, thus, a surface x is a mapping of D into $\mathbb{R}^3$.
This can be denoted by:

$$f : D \to \mathbb{R}^3 \tag{30}$$

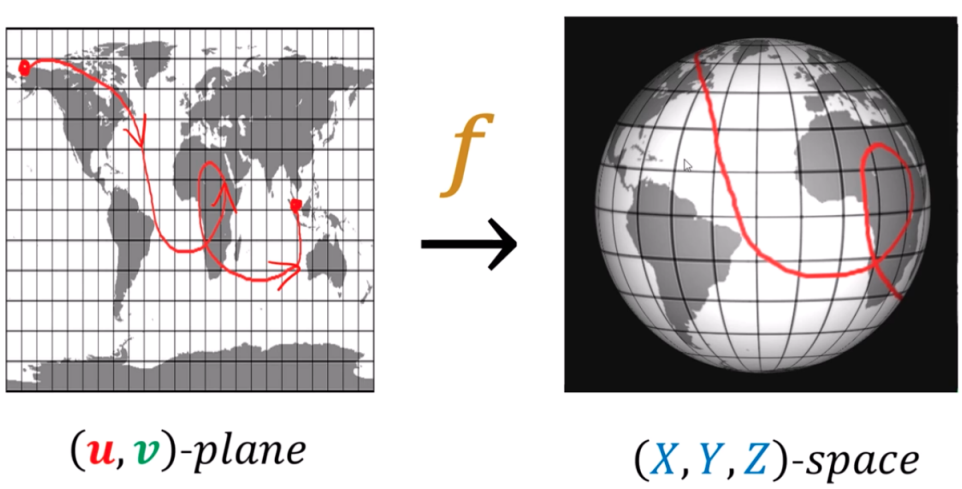
and also represented in 3D coordinate system:

$$(u, v) \to (X(u, v), Y(u, v), Z(u, v))$$

For a sphere surface, its parametrization/transformation function can be written as:

$$\begin{aligned} X &= \cos(v)\sin(u) \\ Y &= \sin(v)\sin(u) \\ Z &= \cos(u) \end{aligned}$$

Allright, and how do we represent a curve on a surface? Well we can apply the same logic, and make the curve, described by γ with parameter t on the uv-plane go through the same transformation function f, to get the curve in the 3D space:



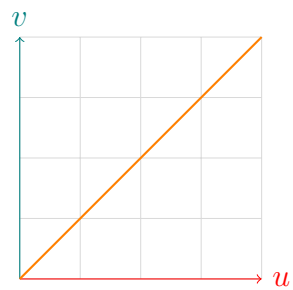
A more mathematically correct definition might be, that given a parametrization for a surface and a curve γ, the curve is on the surface, if a parametrization of γ with a parameter t can be obtained from equations of the surface when u and v become functions of t:

$$\gamma(t) = f(u(t), v(t)) \tag{31}$$

Let's see a simple example of a curve given by:

$$\lambda \rightarrow (u = \lambda, v = \lambda) \tag{32}$$

which is just a simple diagonal line:



Its parametrization on the spherical surface can be obtained by applying the transformation functions we saw:

$$\begin{aligned} X &= \cos(\lambda)\sin(\lambda) \\ Y &= \sin(\lambda)\sin(\lambda) \\ Z &= \cos(\lambda) \end{aligned}$$

If we now want to compute the length of such curve, we need to compute the magnitude of the tangent vector $\frac{dR}{d\lambda}$ where R is just the position vector of a point on the curve, in the 3D space X,Y,Z.

If we try to do that, we obtain:

$$
\begin{aligned}
\left\|\frac{d\vec{R}}{d\lambda}\right\|^2 &= \frac{d\vec{R}}{d\lambda}\cdot\frac{d\vec{R}}{d\lambda} \\
&= \left(\frac{\partial\vec{R}}{\partial X}\frac{dX}{d\lambda}+\frac{\partial\vec{R}}{\partial Y}\frac{dY}{d\lambda}+\frac{\partial\vec{R}}{\partial Z}\frac{dZ}{d\lambda}\right)\cdot\left(\frac{\partial\vec{R}}{\partial X}\frac{dX}{d\lambda}+\frac{\partial\vec{R}}{\partial Y}\frac{dY}{d\lambda}+\frac{\partial\vec{R}}{\partial Z}\frac{dZ}{d\lambda}\right) \\
&= \left(\frac{dX}{d\lambda}\right)^2\left(\frac{\partial\vec{R}}{\partial X}\cdot\frac{\partial\vec{R}}{\partial X}\right)+\frac{dX}{d\lambda}\frac{dY}{d\lambda}\left(\frac{\partial\vec{R}}{\partial X}\cdot\frac{\partial\vec{R}}{\partial Y}\right)+\frac{dX}{d\lambda}\frac{dZ}{d\lambda}\left(\frac{\partial\vec{R}}{\partial X}\cdot\frac{\partial\vec{R}}{\partial Z}\right) \\
&+\frac{dY}{d\lambda}\frac{dX}{d\lambda}\left(\frac{\partial\vec{R}}{\partial Y}\cdot\frac{\partial\vec{R}}{\partial Z}\right)+\left(\frac{dY}{d\lambda}\right)^2\left(\frac{\partial\vec{R}}{\partial Y}\cdot\frac{\partial\vec{R}}{\partial Y}\right)+\frac{dY}{d\lambda}\frac{dZ}{d\lambda}\left(\frac{\partial\vec{R}}{\partial Y}\cdot\frac{\partial\vec{R}}{\partial Z}\right) \\
&+\frac{dZ}{d\lambda}\frac{dX}{d\lambda}\left(\frac{\partial\vec{R}}{\partial Z}\cdot\frac{\partial\vec{R}}{\partial X}\right)+\frac{dZ}{d\lambda}\frac{dY}{d\lambda}\left(\frac{\partial\vec{R}}{\partial Z}\cdot\frac{\partial\vec{R}}{\partial Y}\right)+\left(\frac{dZ}{d\lambda}\right)^2\left(\frac{\partial\vec{R}}{\partial Z}\cdot\frac{\partial\vec{R}}{\partial Z}\right)
\end{aligned}
$$

Due to the orthonormality of the cartesian 3D basis vectors, you would end up with the simple Pythagora's theorem in 3D in this specific case:

$$
\begin{aligned}
\left\|\frac{d\vec{R}}{d\lambda}\right\|^2 &= \frac{d\vec{R}}{d\lambda}\cdot\frac{d\vec{R}}{d\lambda} \\
&= \left(\frac{dX}{d\lambda}\right)^2+\left(\frac{dY}{d\lambda}\right)^2+\left(\frac{dZ}{d\lambda}\right)^2
\end{aligned}
$$

So we can now proceed computing these derivatives of X, Y, Z wrt λ and get:

$$
\begin{aligned}
\frac{dX}{d\lambda} &= \cos(2\lambda) \\
\frac{dY}{d\lambda} &= \sin(2\lambda) \\
\frac{dZ}{d\lambda} &= -\sin(\lambda)
\end{aligned}
$$

$$
\begin{aligned}
\left\|\frac{d\vec{R}}{d\lambda}\right\|^2 &= \frac{d\vec{R}}{d\lambda}\cdot\frac{d\vec{R}}{d\lambda} \\
&= \left(\frac{dX}{d\lambda}\right)^2+\left(\frac{dY}{d\lambda}\right)^2+\left(\frac{dZ}{d\lambda}\right)^2 \\
&= \left(\cos(2\lambda)^2\right)+\left(\sin(2\lambda)^2\right)+\left(-\sin(\lambda)^2\right) \\
&= 1+\sin(\lambda)^2
\end{aligned}
$$

So finally, our arc length is:

$$
\begin{aligned}
\text{arc length} &= \int_0^1 \left\| \frac{d\vec{R}}{d\lambda} \right\| \\
&= \int_0^1 \sqrt{1 + \sin(\lambda)^2} d\lambda \approx 1.12389
\end{aligned}
$$

It's interesting to notice that this is not equal to the length of the same curve in the flat uv-plane, which would just be $\sqrt{2}$.
That big ugly formula we used in the first place, can be rewritten as a series of row array (1 x 3) - matrix (3 x 3) - column array (3 x 1) multiplications.

The 3 x 3 matrix composed by the dot products of the basis vectors, is nothing but the metric tensor in the 3D cartesian coordinate system, which we know actually being indeed the 3 x 3 Identity matrix at the end of the day, due to the orthonormality of the basis.

Something we need to realize, is that to compute this arc length in the curved surface (sphere in this case) we used the XYZ coordinate system, so even though we were analyzing the 2D surface, we then worked in the 3D space around the surface, in order to understand it. This approach of studying surfaces is called **extrinsic** geometry, because it involves analyzing curved 2D spaces, using the sorrounding 3D space that the surface lives in.
The other way of analyzing surfaces is without looking at the sorrounding space, where we would only be looking at the uv-plane, and this point of view is called **intrinsic** geometry.

Using the instrinsic approach, instead of studying the tangent vectors in the XYZ space, we would be studying it in terms of the u and v 2D coordinates:

$$
\frac{d\vec{R}}{d\lambda} = \left(\frac{\partial \vec{R}}{\partial u} \frac{du}{d\lambda} + \frac{\partial \vec{R}}{\partial v} \frac{dv}{d\lambda} \right) \tag{33}
$$

Calculating the magnitude using this tangent vector would give us, in a more compact Einstein notation, considering $u, v = u^1, u^2$:

$$
\begin{aligned}
\left\| \frac{d\vec{R}}{d\lambda} \right\|^2 &= \frac{d\vec{R}}{d\lambda} \cdot \frac{d\vec{R}}{d\lambda} \\
&= \left(\frac{\partial \vec{R}}{\partial u^i} \frac{du^i}{d\lambda} \right) \cdot \left(\frac{\partial \vec{R}}{\partial u^i} \frac{du^j}{d\lambda} \right) \\
&= \frac{du^i}{d\lambda} \frac{du^j}{d\lambda} \left(\frac{\partial \vec{R}}{\partial u^i} \cdot \frac{\partial \vec{R}}{\partial u^j} \right) \\
&= \frac{du^i}{d\lambda} \frac{du^j}{d\lambda} g_{ij}
\end{aligned}
$$

where g_{ij} is the metric tensor of the 2D uv-plane, i.e. the dot products of the basis vectors.

Allright, and how do we get the tangent space basis for the uv-plane? So we can compute the metric tensor...well you remember that:

$$
\begin{aligned}
X &= \cos(v)\sin(u) \\
Y &= \sin(v)\sin(u) \\
Z &= \cos(u)
\end{aligned}
$$

We can expand the tangent basis vectors using the chain rule:

$$
\begin{aligned}
\frac{\partial \vec{R}}{\partial u} &= \frac{\partial X}{\partial u}\frac{\partial \vec{R}}{\partial X} + \frac{\partial Y}{\partial u}\frac{\partial \vec{R}}{\partial Y} + \frac{\partial Z}{\partial u}\frac{\partial \vec{R}}{\partial Z} \\
\frac{\partial \vec{R}}{\partial v} &= \frac{\partial X}{\partial v}\frac{\partial \vec{R}}{\partial X} + \frac{\partial Y}{\partial v}\frac{\partial \vec{R}}{\partial Y} + \frac{\partial Z}{\partial v}\frac{\partial \vec{R}}{\partial Z}
\end{aligned}
$$

Calculating those partial derivatives we get the tangent basis vectors $\vec{e_u}, \vec{e_v}$ in the uv-plane, as linear combination of the 3D basis vectors X, Y, Z:

$$
\begin{aligned}
\vec{e_u} &= \cos(v)\cos(u)\vec{e_x} + \sin(v)\cos(u)\vec{e_y} - \sin(u)\vec{e_z} \\
\vec{e_v} &= -\sin(v)\sin(u)\vec{e_x} + \cos(v)\sin(u)\vec{e_y}
\end{aligned}
$$

And now we're able to compute the metric tensor using the basis vector dot products, and performing the calculations you will end up with:

$$
\begin{bmatrix} \vec{e_u}\cdot\vec{e_u} & \vec{e_u}\cdot\vec{e_v} \\ \vec{e_v}\cdot\vec{e_u} & \vec{e_v}\cdot\vec{e_v} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & \sin(u)^2 \end{bmatrix} \tag{34}
$$

And finally, we can compute the arc length as per the equation above:

$$
\begin{aligned}
\left\| \frac{d\vec{R}}{d\lambda} \right\|^2 &= \frac{d\vec{R}}{d\lambda}\cdot\frac{d\vec{R}}{d\lambda} \\
&= \left(\frac{\partial \vec{R}}{\partial u^i}\frac{du^i}{d\lambda}\right)\cdot\left(\frac{\partial \vec{R}}{\partial u^i}\frac{du^j}{d\lambda}\right) \\
&= \frac{du^i}{d\lambda}\frac{du^j}{d\lambda}\left(\frac{\partial \vec{R}}{\partial u^i}\cdot\frac{\partial \vec{R}}{\partial u^j}\right) \\
&= \frac{du^i}{d\lambda}\frac{du^j}{d\lambda}g_{ij} \\
&= \begin{bmatrix} \frac{du}{d\lambda} & \frac{dv}{d\lambda} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & \sin(u)^2 \end{bmatrix} \begin{bmatrix} \frac{du}{d\lambda} \\ \frac{dv}{d\lambda} \end{bmatrix} \\
&= \left(\frac{du}{d\lambda}\right)^2 + \sin(u)^2\left(\frac{dv}{d\lambda}\right)^2
\end{aligned}
$$

If the arc length we're interested about is given, as we said before, by simple:

$$\lambda \to (u = \lambda, v = \lambda) \tag{35}$$

Then we have that $\frac{du}{d\lambda} = 1$ and $\frac{dv}{d\lambda} = 1$, so we end up with exactly the same result we got when using the extrinsic approach:

$$\begin{aligned} \text{arc length} &= \int_0^1 \left\| \frac{d\vec{R}}{d\lambda} \right\| \\ &= \int_0^1 \sqrt{1 + \sin(\lambda)^2} d\lambda \approx 1.12389 \end{aligned}$$

Main take away from this is that, in curved space, you can't trust your eyeballs, you can only trust the metric tensor

You may wonder one thing, based on the above it seems that in order to get the intrinsic metric tensor matrix, you need to reference the "outside space" anyway. That's actually somewhat true, but there are options to get the intrinsic one from several places:

- Using indeed the extrinsic basis
- Invent one out of imagination
- Obtain it from another equation, like the Einstein Field Equations

Once you have the intrinsic matric, we then can do ALL calculations involving distances and angles, completely intrinsically. But you need that first, and sometimes this involves getting help from the outside space at the start.

The last thing that may still bother you, is the fact that, based on our previous discussions, even in intrinsic geometry we refer to the position vector $\vec{R}$, which references the space outside our curved surface.
If we want to stay completely intrinsic in the surface, we can just remove the position vector from the equations completely, and treat the derivative operators themselves as the vectors

References

[1] eigenchris, *Tensors Calculus*, YouTube video series,
`https://www.youtube.com/playlist?list=PLJHszsWbB6hpk5h8lSfBkVrpjsqvUGTCx`

[2] Lia Vas, *Differential Geometry*, Math 430 lecture notes,
`https://liavas.net/courses/math430/`